

Database Systems Study Guide

Relational Model Basics

A relation is a set of tuples sharing the same attributes. Each attribute draws values from a domain. The degree of a relation is its number of attributes and the cardinality is its number of tuples. Unlike a file, a relation has no inherent ordering of rows.

Keys and Constraints

A superkey is any set of attributes that uniquely identifies a tuple. A candidate key is a minimal superkey. The primary key is the candidate key chosen by the designer. A foreign key references the primary key of another relation and enforces referential integrity.

Functional Dependencies

A functional dependency $X \rightarrow Y$ holds when any two tuples agreeing on X also agree on Y . Armstrong axioms of reflexivity, augmentation and transitivity generate every dependency implied by a given set. The closure of an attribute set is every attribute it determines.

First Normal Form

A relation is in first normal form when every attribute holds only atomic values. Repeating groups and nested relations are not permitted. This is the minimum requirement for any relational table and most systems enforce it structurally.

Second Normal Form

A relation is in second normal form when it is in 1NF and every non-prime attribute is fully functionally dependent on the whole of every candidate key. Partial dependencies on part of a composite key are eliminated at this stage.

Third Normal Form

A relation is in third normal form when it is in 2NF and no non-prime attribute is transitively dependent on any candidate key. A transitive dependency means an attribute depends on another non-prime attribute rather than directly on the key.

Boyce Codd Normal Form

BCNF is stricter than 3NF. Every determinant must be a candidate key. Most relations in 3NF are also in BCNF, but exceptions arise when a relation has multiple overlapping candidate keys. Decomposition into BCNF is not always dependency preserving.

Denormalization

Denormalization deliberately reintroduces redundancy to reduce join cost on read-heavy workloads. It trades write amplification and integrity risk for query speed, and should follow measurement rather than precede it.

B-Tree Indexes

A B-tree index is a balanced tree giving logarithmic lookup. It answers equality predicates and range predicates alike, which is why it is the default index type in most relational engines. Leaf nodes are linked to support ordered scans.

Hash Indexes

A hash index answers equality lookups in near constant time but cannot answer range predicates at all, because hashing destroys ordering. It is rarely the right default despite its lookup advantage.

Composite Indexes

A composite index covers several columns in a defined order. The leftmost prefix rule governs which queries can use it: a query filtering only on the second column cannot use an index whose first column is something else.

Covering Indexes

A covering index contains every column a query needs, so the engine answers from the index alone without visiting the heap. This is often the single largest win available for a hot read path.

Query Planning

The planner estimates the cost of alternative execution strategies using table statistics and picks the cheapest. Stale statistics are a common cause of a plan that was fast last month and is slow today.

Join Algorithms

Nested loop join suits a small outer relation with an indexed inner one. Hash join suits large unsorted inputs on an equality predicate. Merge join suits inputs already sorted on the join column.

Transactions and ACID

Atomicity means all or nothing. Consistency means constraints hold before and after. Isolation means concurrent transactions do not observe each other partially. Durability means a committed change survives a crash.

Isolation Levels

Read uncommitted permits dirty reads. Read committed prevents them but permits non-repeatable reads. Repeatable read prevents those but may permit phantoms. Serializable prevents all three at the highest cost.

Locking and Deadlock

Two-phase locking acquires all locks before releasing any, which guarantees serializability. Deadlock arises when transactions wait on each other cyclically, and is resolved by detection and victim rollback.

Write Ahead Logging

A change is written to the log before it reaches the data pages, so a crash can be recovered by replaying committed work and undoing uncommitted work. This is what makes durability affordable.

Sharding

Sharding partitions rows across independent nodes by a shard key. It scales write throughput but makes cross-shard queries and transactions expensive, so the shard key choice dominates the design.

Replication

Replication copies data to follower nodes for read scaling and failure tolerance. Synchronous replication costs write latency and preserves consistency; asynchronous replication is faster but can lose recent commits on failover.